

Technology acceptance of a VR e-learning application addressing the cellular transport mechanisms

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ABSTRACT

Over the last decades, research has shown that students of all ages experience difficulties in understanding cellular transport mechanisms, especially when it comes to processes on the molecular level such as osmosis and diffusion. One major difficulty in the comprehension of these basic concepts is their abstract nature along with the demand for high spatial ability. To address this problem a virtual reality-based application has been developed, which allows students to freely explore molecular transport mechanisms across the cell membrane by interacting with the environment and manipulating the molecule concentration or initiate transport mechanisms by actively moving molecules. An accompanying study showed an increase in the understanding of osmosis and diffusion after using the e-learning application.

Keywords: Virtual Reality, e-learning, cellular biology, technology acceptance.

1 INTRODUCTION

Over the last decades, research has shown that students of all ages experience difficulties in understanding cellular transport mechanisms, especially when it comes to processes on the molecular level such as osmosis and diffusion [12, 13, 14, 15]. One reason for this lack of understanding is the requirement “to visualize and think about chemical processes at the molecular level” [14], since these processes can't be observed with the naked eye and even are difficult to examine with the help of technological devices such as microscopes [4, 11].

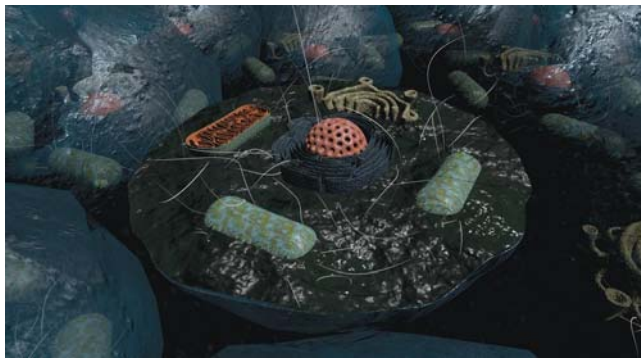


Figure 1: Screenshot of “ViCe”

Commonly used for educational purposes are visualizations in traditional textbooks as well as more advanced digital visualizations such as animations where more dynamic forms of representation, especially when it comes to explain the dynamic nature of processes seem to be superior, compared to static representations [7]. In the past years VR (Virtual Reality) and AR (Augmented Reality) extended the possibilities of e-learning, by offering immersive learning environments. They tend to become more popular since the technical requirements became more efficient and affordable [9]. Interactivity as well as multisensory and three dimensional representation can be considered the major characteristics of a VR [2], where interactivity plays an important role since it extends the learning process to the haptic channel [5]. The use of additional learning channels can result in a better performance in tasks that require spatial ability [9]. The presented study examines two questions: a) Do the affordances of a VR help students with their understanding of the basic cellular transport mechanisms and b) is this form of representation accepted by them.

2 METHODS

To address the previously mentioned misconceptions held by students, a VR application (ViCe “Virtual Cell”, see fig. 1) has been developed. Since the successful development of innovative interventions is strongly connected with the involvement of learners, teachers and developers the application has been developed in iterative cycles of design, evaluation and redesign, according to methods of the design-based research [1]. For testing the students’ learning achievements a pre-/ posttest, based on the ODCA (“Osmosis and Diffusion Conceptual Assessment”) [6] has been conducted with the final version of “ViCe”. The technology acceptance has been evaluated directly after the posttest with a Likert-scaled questionnaire that has been introduced by Holden and Rada [8] who refined the more commonly used UTAUT (“Unified Theory of Acceptance and Use of Technology”) [16], by adding the usability component.

3 RESULTS

15 biology students ($N = 15$, at the age of 19–37, $M = 23.6$, $SD = 5.3$) participated. Initially the participants accomplished the ODCA. After a standardized introduction to the VR-system and its components (HTC Vive) the intervention took about 12 minutes per participant. The posttest showed that the participants could increase their performance on average by about 1 point, compared to the pretest. On average the participants achieved a result of 11.87 ($SD = 3.18$) out of 18 possible points on the the ODCA pretest. After the intervention the performance could be enhanced to an average of 12.80 points ($SD = 2.68$) (see fig. 2).

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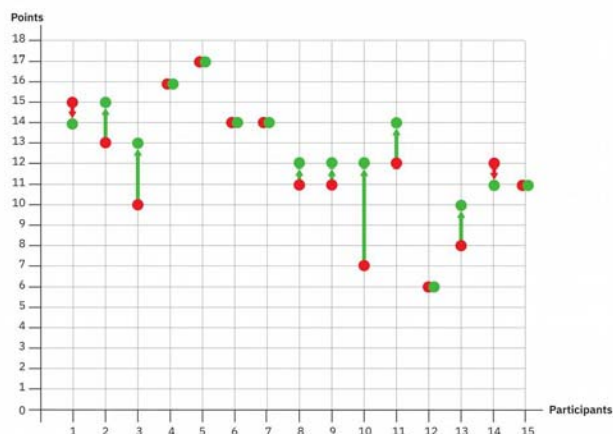


Figure 2: Comparison of pre-/ posttest ODCA

The data corresponded to a normal distribution so that a paired t-test could be calculated ($t(14) = 2.23$, $p = .043$ n). Due to the small sample size, the results should be considered preliminary [3]. It is planned to increase the sample size in the coming months. After the posttest the participants answered the questionnaire on the technology acceptance. The items for the “Perceived Usefulness + Perceived Ease of Use + Usability” gave a mean value of $M = 3.49$ by mapping the Likert-scale to values from 0–4 (0 = Strongly Disagree, 1 = Disagree, 2 = Neutral, 3 = Agree, 4 = Strongly Agree). The items for “Perceived Ease of Use + Usability” yielded a result of $M = 3.48$ and the items measuring only the “Perceived Usefulness” $M = 3.44$. Combined this leads to a result of $M = 3.47$ what can be considered a high acceptance (see fig. 3).

Subject	PU + PEU + U					PEU + U					PU					Combined		
	I01	I02	I03	I04	I05	I06	I07	I08	I09	I10	I11	I12	I13	I14	I01-05	I06-09	I10-14	
01	4	3	4	4	3	3	4	2	4	3	4	3	4	3	3.6	3.25	3.4	
02	3	2	3	3	2	3	4	4	3	3	4	3	3	3	2.6	3.5	3	
03	4	3	4	4	4	4	4	4	4	4	4	4	4	4	3.8	4	4	
04	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	
05	4	3	2	3	4	4	4	2	4	3	4	4	4	4	3.2	3.5	3.8	
06	4	4	4	4	3	3	4	2	4	2	3	3	3	3	3.8	3.25	2.8	
07	3	3	4	4	4	4	3	3	3	4	4	4	4	3	3.6	3.5	3.6	
08	4	3	4	4	3	4	3	3	4	3	3	3	4	3	3.6	3.5	3.2	
09	4	3	4	4	3	4	4	4	4	4	3	4	4	3	3.6	4	3.4	
10	4	3	4	4	4	3	4	2	4	4	4	4	3	4	3.8	3.25	3.8	
11	4	4	3	2	3	3	4	2	3	3	3	3	4	3	3.2	3	3.2	
12	4	4	4	4	3	3	4	4	4	4	3	3	3	3	3.8	3.75	2.8	
13	3	4	3	2	3	4	3	3	3	3	4	4	3	3	3	3.25	3.4	
14	4	4	3	3	4	4	2	3	3	3	4	3	4	4	3.6	3	3.6	
15	3	4	4	3	2	4	4	3	3	3	3	4	4	4	3.2	3.5	3.6	
Means	3.73	3.40	3.60	3.47	3.27	3.60	3.73	3.00	3.60	3.20	3.67	3.40	3.67	3.27	3.49	3.48	3.44	

Figure 3: Evaluation of technology acceptance with results for every item (I01– I14).

4 DISCUSSION

This study on the usage of a VR e-learning application in the field of cellular biology, once more shows the efficiency of design-based research, when it comes to the development of innovative interventions. Involving experts of the subject didactics as well as users, proved to be the right choice in reference to the results of the technology acceptance evaluation. Considering the relatively short time of about 12 minutes the intervention takes a participant to accomplish, the improved performance on the ODCA also shows its high potential.

To validate the results and prove the efficiency of the intervention, further research will be conducted. Simultaneously the results will be incorporated in a further cycle of redesign as promoted by the iterative nature of the design-based research.

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